

Software Requirements Specification

(SRS)

Project: Online Bookstore

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Version	Date	Author	Change Summary	Approval
1.0	04-09-2026	Team 4	Initial SRS prepared for the Online Bookstore project	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Mentor			

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1. Introduction

1.1 Purpose

This document describes the requirements for an Online Bookstore developed as a web-based application. The system is intended to make the process of discovering, purchasing and managing books simpler for customers, while giving administrators a practical way to maintain the catalogue, inventory and orders. The SRS records the functional behaviour of the system, quality expectations, interfaces and test conditions so that the team has a common reference during development and testing.

1.2 Scope

The project covers customer registration and login, browsing and searching books, viewing book details, managing a shopping cart, placing orders, making payments through an integrated payment service, tracking orders and submitting ratings or reviews. The administrative side covers managing books, categories, stock, users, orders, reviews and basic sales reports. The project does not include book publishing workflows, physical warehouse operations or direct integration with every courier provider.

1.3 Audience

Developers, testers, project mentor, course evaluators, administrators and future maintainers of the application.

1.4 Definitions

- User/Customer: A registered or guest visitor who browses and purchases books.
- Administrator: An authorised user responsible for managing catalogue, inventory, users and orders.
- Cart: A temporary collection of books selected by a customer before checkout.
- Order: A confirmed purchase containing one or more books and delivery details.
- API: Application Programming Interface used for communication between the frontend, backend and external services.
- JWT: JSON Web Token used for maintaining authenticated user sessions.
- MERN: MongoDB, Express.js, React and Node.js.
- Django: Python web framework that may be used for backend services or modules.

2. Overall Description

2.1 Product Perspective

The Online Bookstore is a standalone web application with a customer-facing interface and an administrative interface. A typical deployment uses a React-based frontend communicating with a REST API implemented using Node.js/Express

or Django. The application stores user, catalogue, cart and order data in a database and may communicate with external payment and notification services.

2.2 Major Product Functions

- Account registration, login and logout.
- Browse books by category, author or title.
- Search and filter the catalogue.
- Display detailed information such as author, description, price and availability.
- Add, update and remove books in the shopping cart.
- Manage wishlist items.
- Checkout with address and payment information.
- Create, view, track and cancel eligible orders.
- Submit ratings and reviews for purchased books.
- Administrative management of books, categories and inventory.
- Administrative order and user management.
- Review moderation and sales reporting.

2.3 User Roles and Characteristics

- Customer: Uses the website to discover and purchase books. The interface should be simple enough for users with ordinary web and mobile experience.
- Guest: Can browse and search the catalogue but must create an account or log in before completing actions that require an account, such as order history and reviews.
- Administrator: Has authorised access to catalogue, inventory, user and order management functions and is expected to understand the basic administrative workflow.

2.4 Operating Environment

Modern desktop and mobile web browsers, internet connection, a web server/application server, database server and HTTPS-enabled deployment environment.

2.5 Constraints

- The application must use the selected MERN/Django-compatible technology stack.
- Payment processing must rely on an approved external payment gateway or sandbox during development.
- Passwords must not be stored in plain text.
- The design must support both desktop and common mobile screen sizes.
- Third-party service availability can affect payment or notification-related functions.

3. External Interface Requirements

3.1 User Interfaces

The customer interface shall provide a clear navigation bar, search functionality, category browsing, book cards, a book detail page, cart and checkout pages, and an account/order section. The administrator interface shall provide protected pages for catalogue, inventory, order, user and review management. Forms should show understandable validation messages instead of exposing technical errors.

3.2 Hardware Interfaces

The application does not require specialised hardware. It shall operate on standard computers, laptops, tablets and smartphones with supported web browsers.

3.3 Software Interfaces

- Frontend web application communicating with backend services through REST APIs.
- Database for users, books, categories, carts, orders, reviews and related records.
- Authentication service using JWT or an equivalent secure session mechanism.
- Payment gateway API for payment authorisation and confirmation.
- Optional email/notification service for order updates and account-related messages.

3.4 Communications

Client-server communication shall use HTTPS in deployed environments. API requests and responses shall use JSON. The system shall handle failed external requests gracefully and show the user an understandable status instead of leaving the order in an unclear state.

4. System Features (Detailed)

The following functional requirements use the identifier format OBS-F-###. Each requirement includes an acceptance condition and a reference test case.

4.1 Account and Authentication

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-001	The system shall allow a new user to register using a unique email address and a password that	Functional	High	Customer	AC: Valid registration creates an account and confirmation is shown. Test: TC-AUTH-01	Depends on user database and validation rules.

	meets the configured validation rules.					
OBS-F-002	The system shall authenticate registered users using their email and password and create an authorised session after successful verification.	Functional	High	Customer	AC: Valid credentials allow access to account features. Test: TC-AUTH-02	Requires secure password verification.
OBS-F-003	The system shall allow an authenticated user to log out and end the active session.	Functional	Medium	Customer	AC: Protected pages are no longer accessible through the active session. Test: TC-AUTH-03	Requires session/JWT handling.

4.2 Browse, Search and Book Details

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-004	The system shall display available books with title, author, price, cover image and availability information.	Functional	High	Customer	AC: Catalogue data is visible for all active books. Test: TC-CAT-01	Requires catalogue data.
OBS-F-005	The system shall allow users to search books by title, author or relevant keyword.	Functional	High	Customer	AC: Matching results are returned for a valid search term. Test: TC-SRCH-01	Search depends on indexed/available catalogue data.
OBS-F-006	The system shall allow users to filter or browse books by category and other supported criteria such as price range or availability.	Functional	Medium	Customer	AC: Results update according to selected filters. Test: TC-SRCH-02	Filter options depend on available metadata.

OBS-F-007	The system shall display a detailed page for a selected book including description, author, category, price,	Functional	High	Customer	AC: Selected book details match stored catalogue data. Test: TC-BOOK-01	Requires book and review data.
	stock status and user ratings where available.					

4.3 Cart and Wishlist Management

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-008	The system shall allow a customer to add an available book to the shopping cart.	Functional	High	Customer	AC: Selected book appears in the cart with correct quantity and price. Test: TC-CART-01	Requires stock availability check.
OBS-F-009	The system shall allow a customer to update quantities or remove items from the shopping cart.	Functional	High	Customer	AC: Cart total and item count update correctly. Test: TC-CART-02	Quantity cannot exceed available stock.
OBS-F-010	The system shall allow authenticated customers to add and remove books from a personal wishlist.	Functional	Medium	Customer	AC: Wishlist reflects add/remove actions after refresh. Test: TC-WISH-01	Requires authenticated user account.

4.4 Checkout and Order Management

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-011	The system shall collect delivery information and show an order summary before payment confirmation.	Functional	High	Customer	AC: Customer can review items, total and delivery details before placing the order. Test: TC-CHK-01	Requires non-empty cart and valid address fields.

OBS-F-012	The system shall initiate payment through the configured payment service and create an order only after a successful or explicitly supported payment confirmation.	Functional	High	Customer / Business	AC: Successful payment creates one order with the correct total. Test: TC-PAY-01	Depends on payment gateway/sandbox.
OBS-F-013	The system shall allow customers to view their order history and current order status.	Functional	High	Customer	AC: Customer sees only their own orders with correct status. Test: TC-ORD-01	Requires order records and access control.
OBS-F-014	The system shall allow customers to cancel an order only when its status is within the configured cancellation window.	Functional	Medium	Customer / Business	AC: Eligible order is cancelled; ineligible order shows a reason. Test: TC-ORD-02	Cancellation rules must be configured.

4.5 Reviews and Administration

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-015	The system shall allow an eligible customer to submit a rating and review	Functional	Medium	Customer	AC: Valid review is stored and associated with the correct book and	Eligibility may be based on completed/confirmed purchase.
	for a purchased book.				user. Test: TC-REV-01	
OBS-F-016	The system shall allow authorised administrators to add, edit, activate or deactivate book records and manage categories.	Functional	High	Administrator	AC: Catalogue changes are visible according to their active status. Test: TC-ADM-01	Requires admin role check.
OBS-F-017	The system shall allow authorised administrators to update inventory and manage order status changes.	Functional	High	Administrator	AC: Stock and order status updates are persisted and reflected to relevant users. Test: TC-ADM-02	Requires admin role and order workflow rules.

OBS-F-018	The system shall allow authorised administrators to moderate inappropriate reviews and view basic sales/order information.	Functional	Medium	Administrator	AC: Moderation changes review visibility and reports show stored order data. Test: TC-ADM-03	Requires reporting queries and moderation status.
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5. Non-functional Requirements (Detailed)

The following non-functional requirements are measurable and use the identifier format OBS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
OBS-NF-001	Under normal load, the catalogue and book detail pages shall respond within 3 seconds for at least 90% of tested requests.	Performance	High	90th percentile response time ≤ 3 seconds. Test: TC-PERF-01.
OBS-NF-002	The application shall maintain 99.5% monthly availability excluding planned maintenance and unavoidable third-party outages.	Reliability	High	Availability monitoring/reporting shows $\geq 99.5\%$. Test: TC-REL-01.
OBS-NF-003	The customer-facing interface shall remain usable on common desktop and mobile screen sizes supported by the project.	Usability / Compatibility	High	Key flows pass responsive testing on representative desktop and mobile viewports. Test: TC-UX-01.
OBS-NF-004	The system shall preserve order and payment-related records so that a failed or interrupted checkout can be investigated without silently losing confirmed transactions.	Data Integrity	High	Tested failure scenarios retain auditable order/payment status. Test: TC-DATA-01.

OBS-NF-005	The application shall provide clear validation and error messages for invalid input and failed user actions without exposing internal stack traces or sensitive implementation details.	Usability / Security	Medium	Invalid input tests show understandable messages and no internal error details. Test: TC-ERR-01.
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5.1 Security

5.1.1 Security Objectives

- SO-01 – Protect user accounts and personal information by ensuring that only authorised users can access protected data and functions.
- SO-02 – Protect the integrity of orders, payments and administrative operations so that unauthorised users cannot alter business-critical records.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance criteria
OBS-SR-001	The system shall store user passwords only as salted cryptographic hashes and shall never store or display them in plain text.	Security	High	Password storage inspection confirms no plaintext passwords. Test: TC-SEC-01.
OBS-SR-002	The system shall enforce role-based access control so that administrative functions are inaccessible to normal customers.	Security	High	Customer attempts to access admin APIs/pages are rejected. Test: TC-SEC-02.
OBS-SR-003	The system shall validate and sanitise server-side input used in account, search, review, cart and administrative operations.	Security	High	Representative invalid/malicious input tests are rejected safely. Test: TC-SEC-03.
OBS-SR-004	The system shall use HTTPS for authenticated and sensitive client-server communication in deployed environments.	Security	High	Deployment configuration redirects/protects supported sensitive endpoints over HTTPS. Test: TC-SEC-04.

OBS-SR-005	The system shall record security-relevant administrative and authentication events without storing passwords or other secrets in application logs.	Security	Medium	Log review confirms required events and absence of secrets. Test: TC-SEC-05.
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6. Quality Attributes & Acceptance Tests

The project will be accepted when all high-priority functional requirements have been implemented and verified, all critical security requirements have passed their planned tests, and no unresolved defect prevents a customer from completing the main browse-to-order flow.

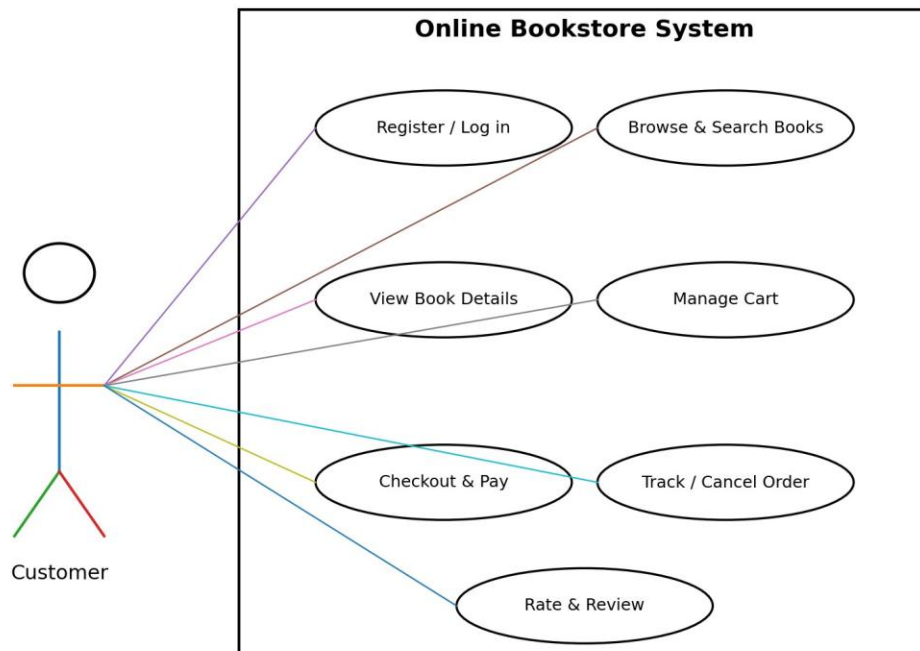
Acceptance Test Suites

- Authentication tests: registration, login, logout and protected access.
- Catalogue tests: browsing, searching, filtering and book details.
- Cart and wishlist tests: add, update, remove and quantity validation.
- Checkout and order tests: address validation, payment outcome handling, order creation, history and cancellation rules.
- Administrative tests: catalogue, inventory, category, order and review management.
- Performance and responsiveness tests.
- Security tests: password protection, role access, input validation, HTTPS configuration and safe logging.

7. System Models and UML Use-Case Diagrams

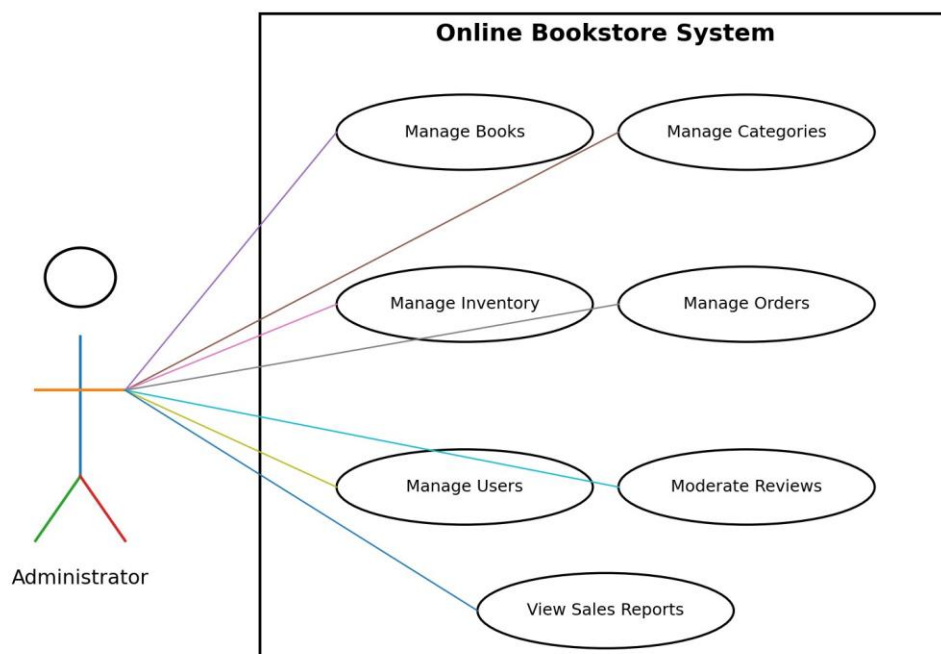
7.1 Customer Use-Case Diagram

This diagram summarises the main interactions between a customer and the Online Bookstore. The customer can discover books, manage a cart, complete checkout, manage orders and submit reviews.



7.2 Administrator Use-Case Diagram

This diagram summarises the functions available to an authorised administrator. Administrative responsibilities focus on maintaining the catalogue and inventory, handling orders and users, moderating reviews and reviewing basic sales information.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
OBS-F-001	Register account	4.1 / DS-AUTH-01	Auth Module	TC-AUTH-01	N	
OBS-F-002	Login user	4.1 / DS-AUTH-02	Auth Module	TC-AUTH-02	N	
OBS-F-004	Browse catalogue	4.2 / DS-CAT-01	Catalogue Module	TC-CAT-01	N	
OBS-F-005	Search books	4.2 / DS-SRCH-01	Search Module	TC-SRCH-01	N	
OBS-F-008	Add to cart	4.3 / DS-CART-01	Cart Module	TC-CART-01	N	
OBS-F-011	Checkout summary	4.4 / DS-CHK-01	Checkout Module	TC-CHK-01	N	
OBS-F-012	Payment and order creation	4.4 / DS-PAY-01	Order/Payment Module	TC-PAY-01	N	
OBS-F-013	View order history	4.4 / DS-ORD-01	Order Module	TC-ORD-01	N	
OBS-F-015	Submit review	4.5 / DS-REV-01	Review Module	TC-REV-01	N	
OBS-F-016	Manage books	4.5 / DS-ADM-01	Admin Module	TC-ADM-01	N	
OBS-NF-001	Response time target	5 / DS-PERF-01	Frontend/API	TC-PERF-01	N	
OBS-NF-004	Transaction data integrity	5 / DS-DATA-01	Order/Payment Module	TC-DATA-01	N	
OBS-SR-002	Role-based access	5.1 / DS-SEC-02	Auth/Admin Module	TC-SEC-02	N	
OBS-SR-003	Input validation	5.1 / DS-SEC-03	API Layer	TC-SEC-03	N	